

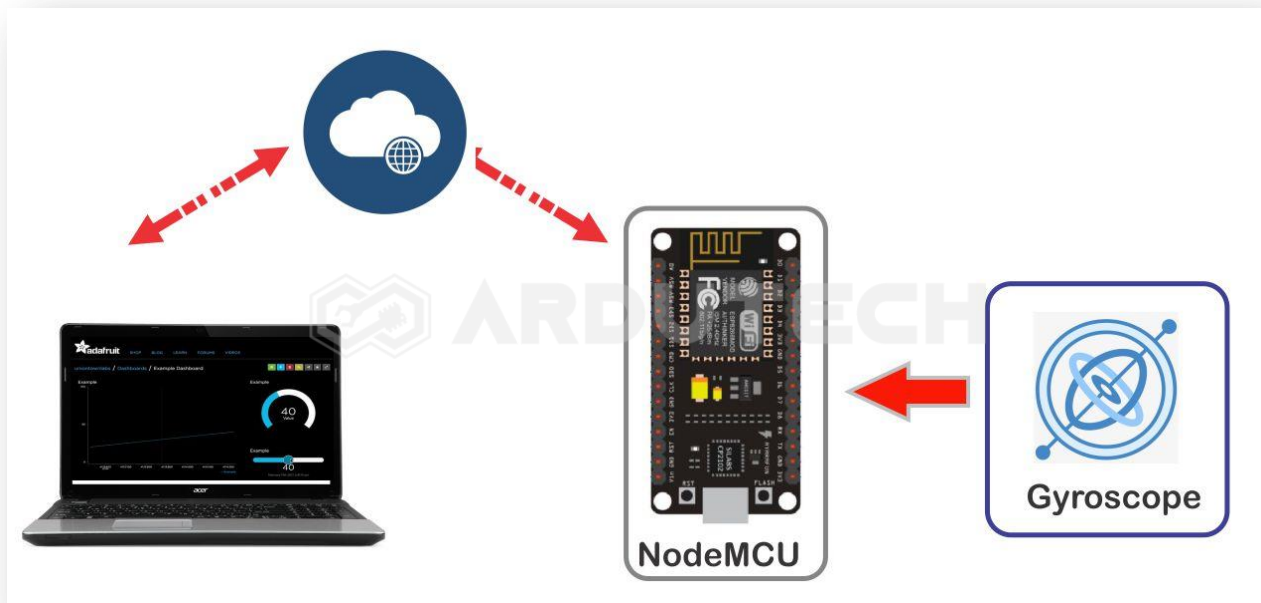
IoT Data Logger Suhu dan Accelerometer

Deskripsi

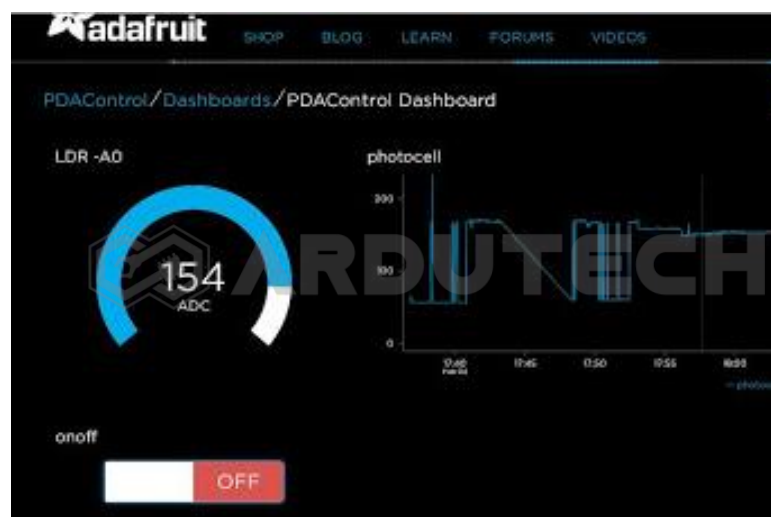
Membuat proyek IoT (*Internet of Things*) untuk membaca data sensor suhu dan accelerometer MPU-6050 kemudian data dikirim secara kontinyu ke Adafruit IO dengan protokol MQTT.

Cara Kerja

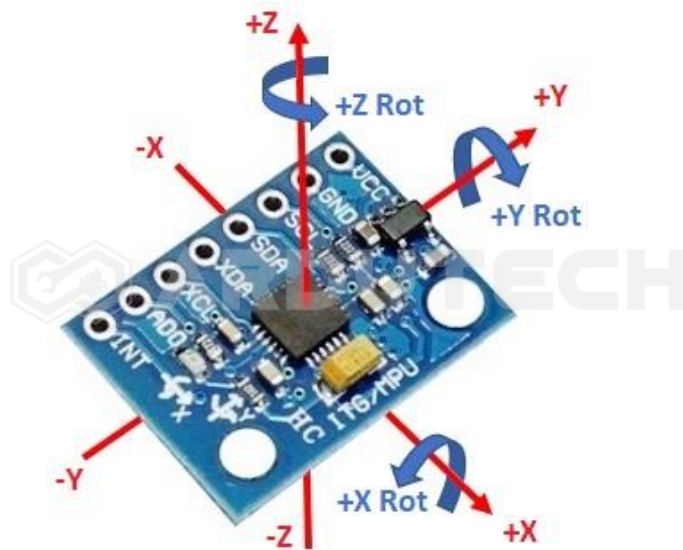
Sensor modul MPU-6050 (accelerometer & gyroscope sensor) membaca parameter berupa percepatan suatu obyek yang bergerak, orientasi gerak sebuah obyek beserta temperatur kemudian hasilnya dikirimkan ke NodeMCU. Data kemudian diolah dan oleh NodeMCU yang terhubung dengan jaringan internet (WiFi) kemudian akan mengirimkan ke server Adafruit IO dengan antarmuka (GUI) yang kita buat secara custom.



Adafruit IO adalah sistem yang disediakan oleh Adafruit (pembuat modul – modul sensor, perangkat elektronik, robot dll) untuk membuat sebuah proyek/aplikasi IoT (*Internet of Things*). Adafruit IO ini boleh dipakai oleh siapa saja karena sifatnya free.



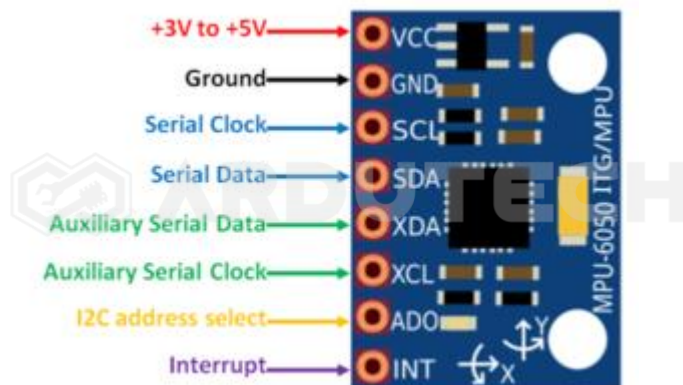
Modul sensor MPU-6050 merupakan sensor accelerometer dan gyroscope. Accelerometer dipakai untuk mengukur percepatan obyek yang bergerak baik dinamis maupun statis. Pengukuran dinamis adalah pengukuran percepatan pada objek yang bergerak sedangkan pengukuran statis adalah pengukuran terhadap gravitasi bumi.



MPU-6050 sudah dikemas dalam sebuah modul seperti GY-521 yang tentu saja menggunakan chip MPU-6050 (3-axis accelerometer dan 3-axis gyroscope) dari IvenSense. Modul ini berkomunikasi dengan sistem I2C seperti terlihat pada jalur koneksi terdapat pin SCL dan SDA.

Spesifikasi :

- Chipset: MPU-5060
- Tegangan 3.3-5V DC
- Jalur komunikasi: I2C up to 400kHz
- Range Gyroscope: 250, 500, 1000, 2000/s
- Range Acceleration: 2, 4, 6, 8, 16g
- Chip built-in 16bit AD converter, 16-bit data output



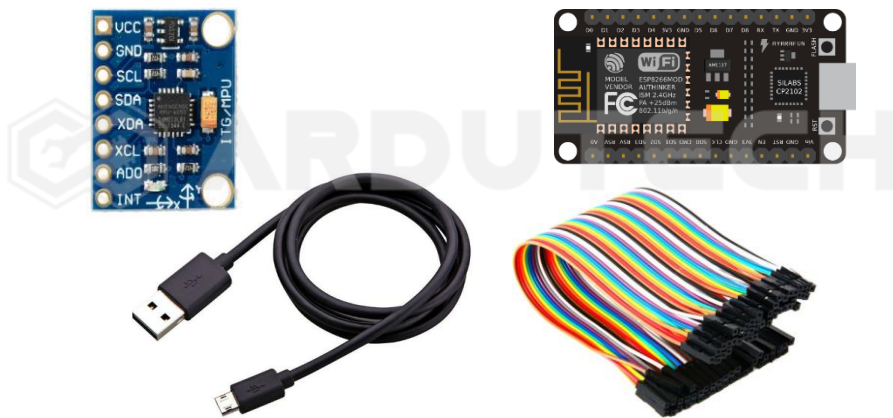
Keterangan pin/kaki.

- VCC : Power 3 – 5 V
- GND : Ground
- SCL : Serial Clock pin. Terhubung ke pin SCL microcontroller

- SDA : Serial Data pin. Terhubung ke pin SDA microcontroller
- XCL : Auxiliary Serial Clock pin. Terhubung ke I2C (SCL) yang lain.
- XDA : Auxiliary Serial Data pin. Terhubung ke I2C (SDA) yang lain.
- AD0 : I2C Slave Address LSB
- INT : Pin interupsi

Kebutuhan Bahan

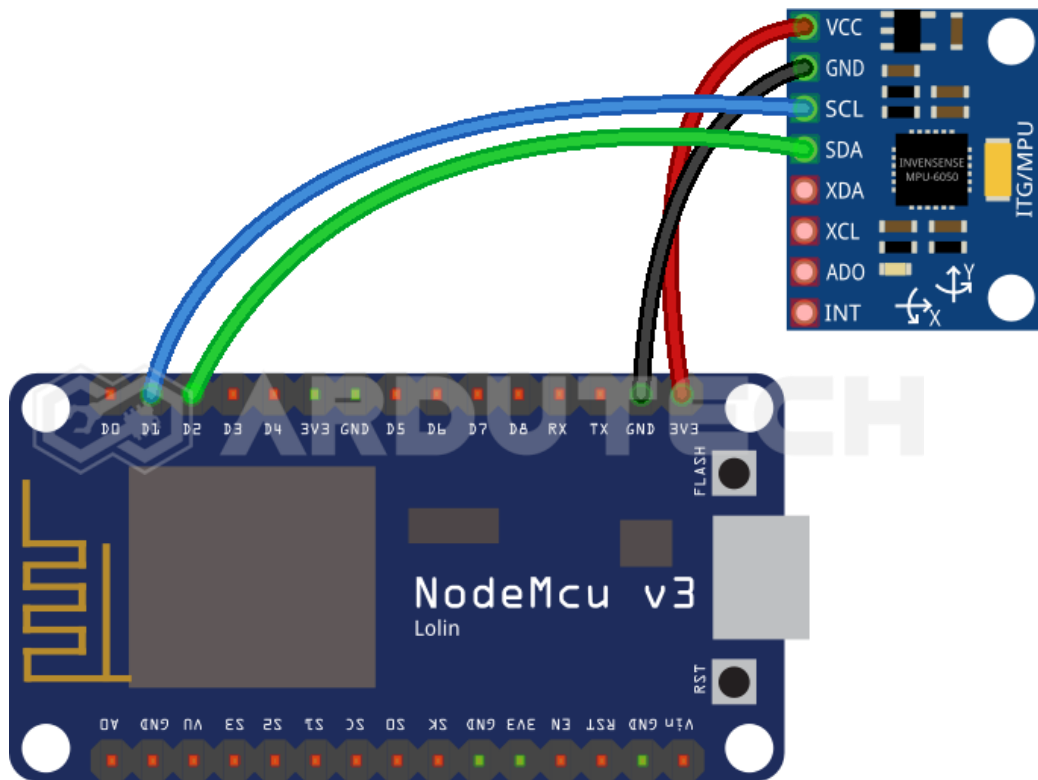
- NodeMCU V3
- Modul MPU-6050
- Kabel konektor
- Kabel micro USB



Kebutuhan Software

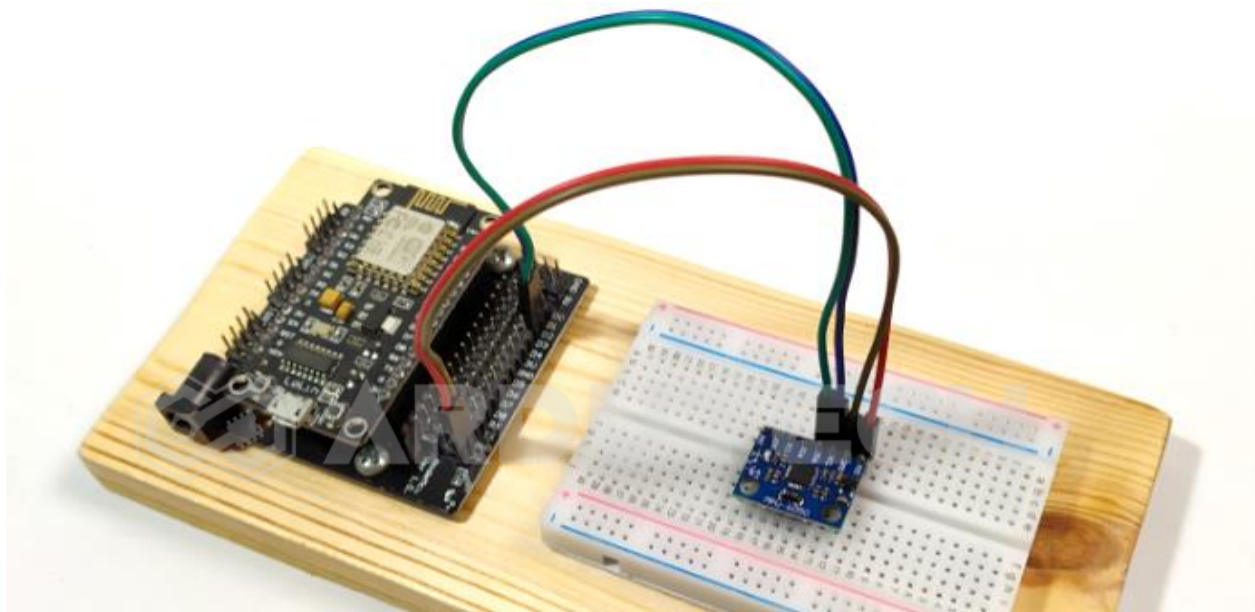
- Arduino IDE
- Blynk

Rangkaian/Skematik



Koneksi NodeMCU dengan Modul MPU-6050 :

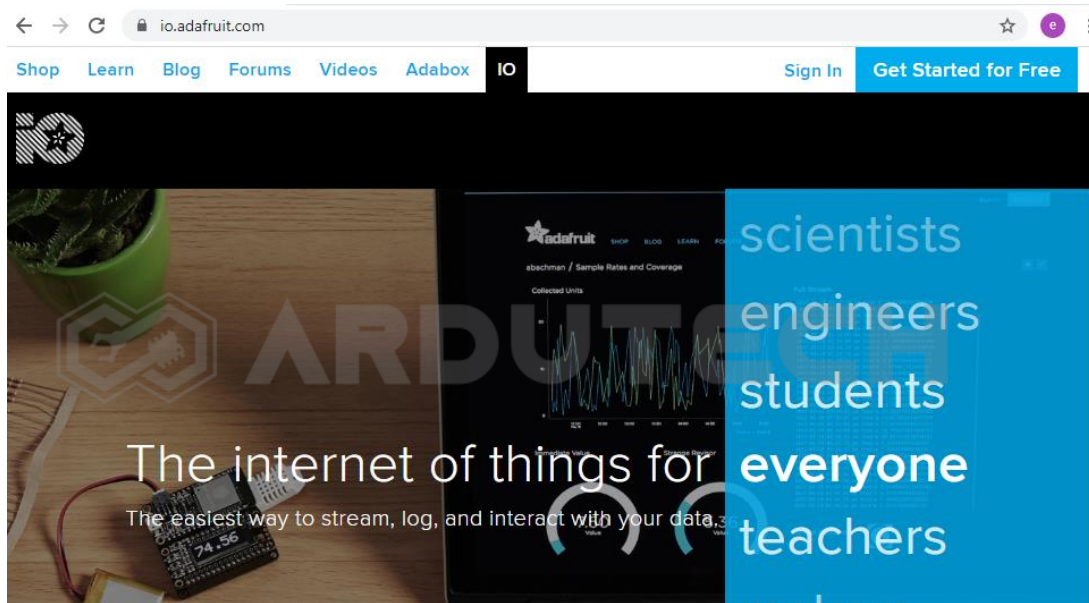
NodeMCU	MPU-6050
D1	SCL
D2	SDA
GND	GND
3V3	VCC



Membuat Antarmuka di Adafruit

Silakan masuk ke halaman <https://io.adafruit.com/>

www.ardutech.com @2020 (Terimakasih anda tidak meng-copy file ini untuk dijual kembali)



Jika belum mempunyai akun silakan membuat akun terlebih dahulu. Siapkan sebuah akun email aktif.

Klik pada **“Get Started for free”** di bagian pojok kanan atas.

SIGN UP

The best way to shop with Adafruit is to create an account which allows you to shop faster, track the status of your current orders, review your previous orders and take advantage of our other member benefits.

FIRST NAME

LAST NAME

EMAIL

USERNAME

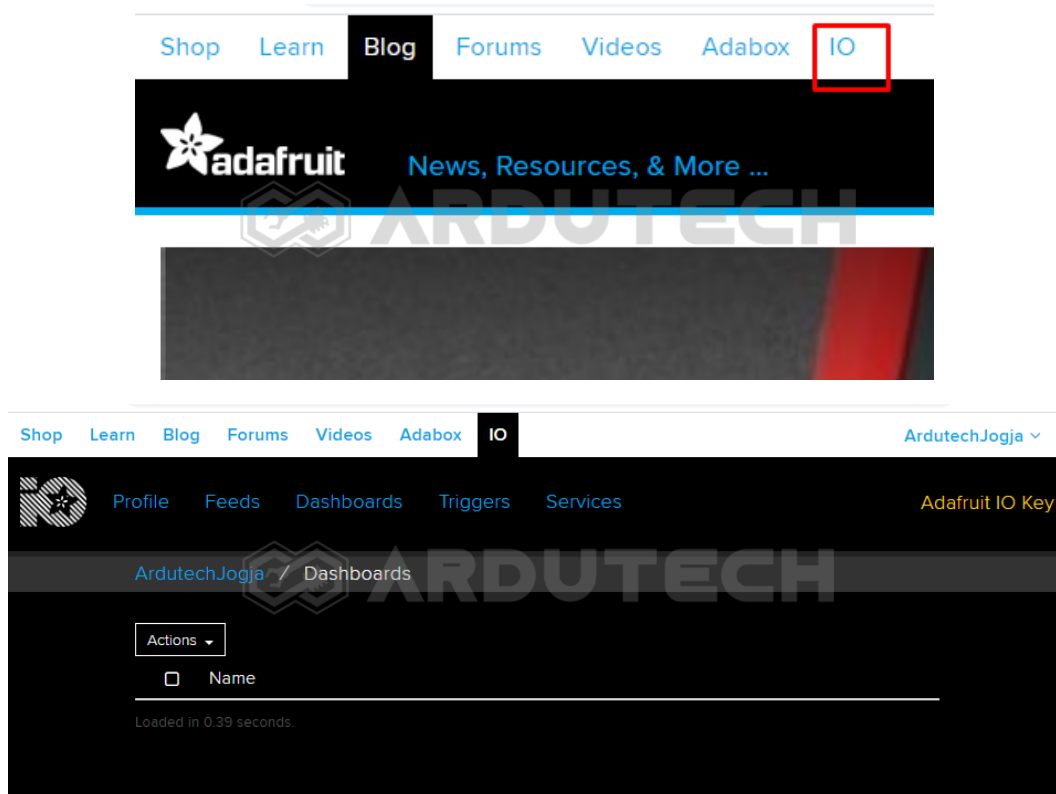
Username is viewable to the public on the forums, Adafruit IO, and elsewhere.

PASSWORD

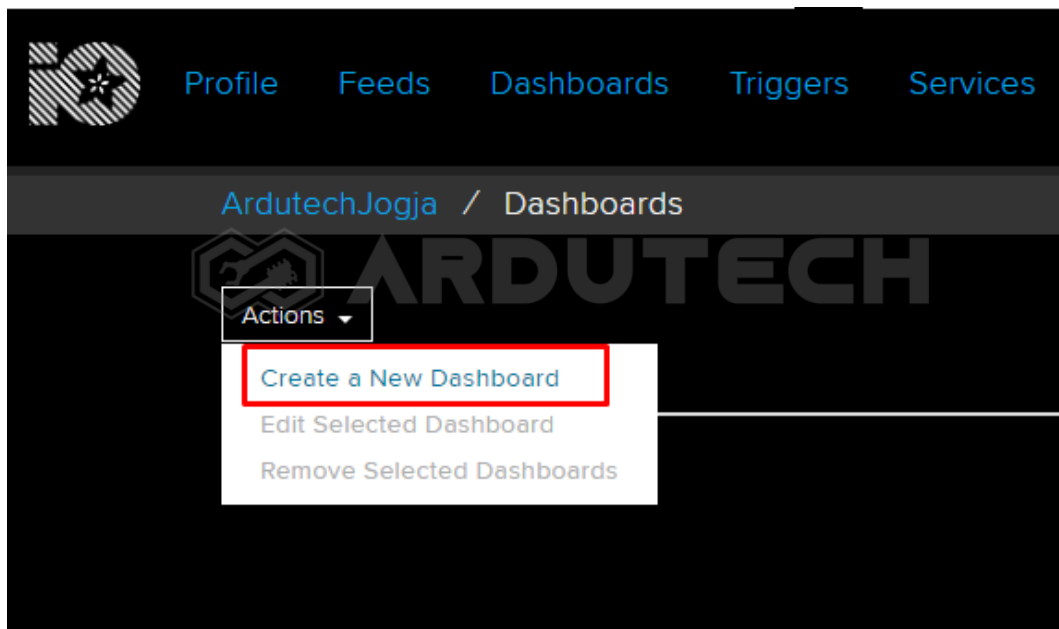
CREATE ACCOUNT

Isi data – data yang diperlukan kemudian klik **“CREATE ACCOUNT”** dan selanjutnya ikuti petunjuknya.

Masuk ke menu **IO**.



Klik pada tombol “Actions” kemudian pilih “Create a New Dashboard”



Selanjutnya akan muncul jendela untuk memberi nama, silakan beri nama pada dashboard yang akan dibuat, misalnya “IoT Datalogger Temperature dan Accelerometer” dan klik “Create”.



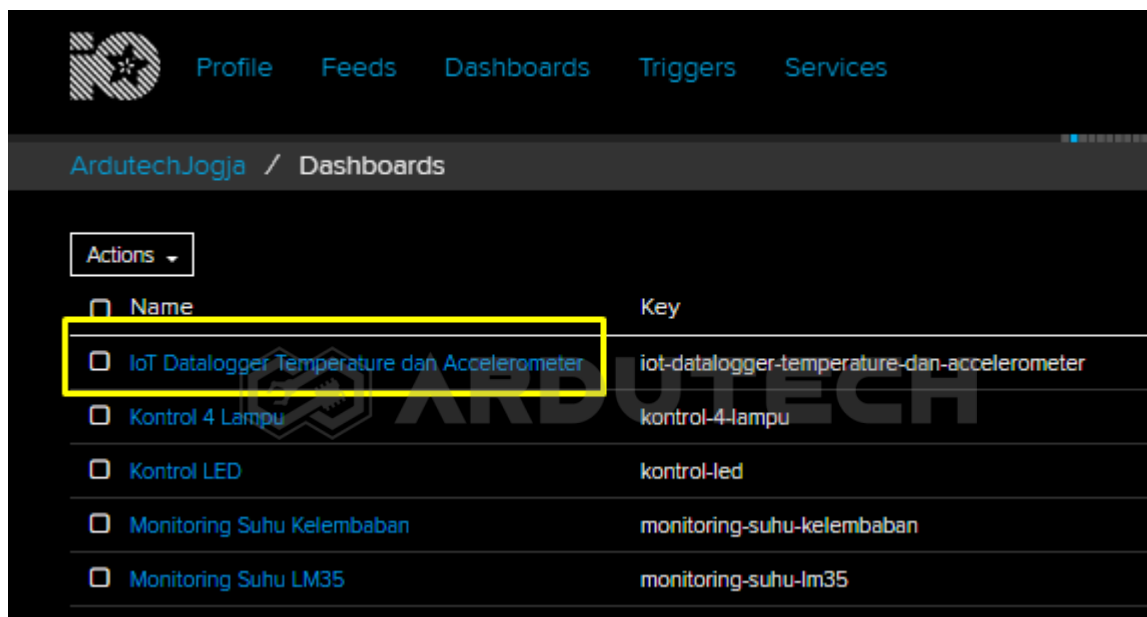
Create a new Dashboard

Name
IoT Datalogger Temperature dan Accelerometer

Description

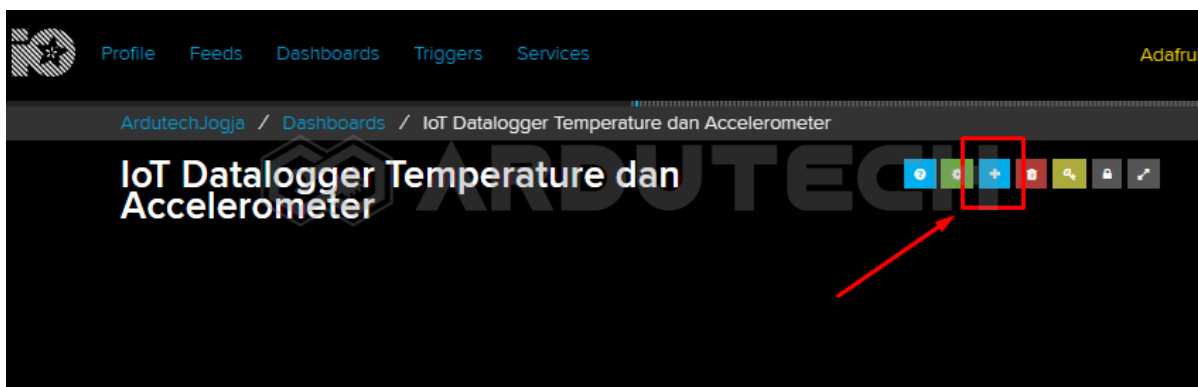
Cancel Create

Setelah klik tombol **“Create”** , akan muncul dashboard baru dengan nama **“IoT Datalogger Temperature dan Accelerometer”**.



Name	Key
☐ IoT Datalogger Temperature dan Accelerometer	iot-datalogger-temperature-dan-accelerometer
☐ Kontrol 4 Lampu	kontrol-4-lampu
☐ Kontrol LED	kontrol-led
☐ Monitoring Suhu Kelembaban	monitoring-suhu-kelembaban
☐ Monitoring Suhu LM35	monitoring-suhu-lm35

Klik **“IoT Datalogger Temperature dan Accelerometer”** kemudian pilih tombol **“+”** untuk membuat blok baru.



Setelah tombol **“+”** di-klik akan muncul tampilan bermacam ‘widget’. Ada tombol (Toggle), tampilan Gauge, Slider, Text dll. Pilih **Gauge** (pojok kanan atas).

Create a new block

Click on the block you would like to add to your dashboard. You can always come back and switch the block type later if you change your mind.



Selanjutnya muncul tampilan untuk memilih 'Chose feed'. Ini semacam "Topik" dalam komunikasi MQTT. Jadi supaya dikenali oleh 'subscriber' (penerima perintah) maka 'publisher' harus menentukan 'topik' nya. Silakan tulis **"temperature"** kemudian klik tombol **"Create"**.

Choose feed

Gauge: A gauge is a read only block type that shows a fixed range of values.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Search field:

Group / Feed	Last value	Recorded
--------------	------------	----------

Muncul feed temperature

Choose feed

Gauge: A gauge is a read only block type that shows a fixed range of values.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Search field:

Group / Feed	Last value	Recorded
☐ LED	ON3	about 12 hours
☐ Light	0	about 13 hours
☐ photocell	561	about 12 hours
☒ temperature	33.55	2 minutes

Selanjutnya klik **"Next Step >"** sehingga muncul tampilan **Block settings** :

Block settings

In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

Block Title (optional)

Temperature

Gauge Min Value

0

Gauge Max Value

100

Gauge Width

50px

Gauge Label

Celsius

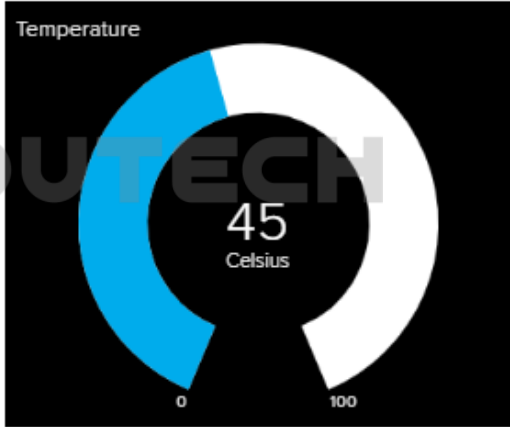
Low Warning Value

Optional. If no low warning value is given, the gauge will only change color when the value is out of bounds.

High Warning Value

Block Preview

Temperature



Gauge A gauge is a read only block type that shows a fixed range of values.

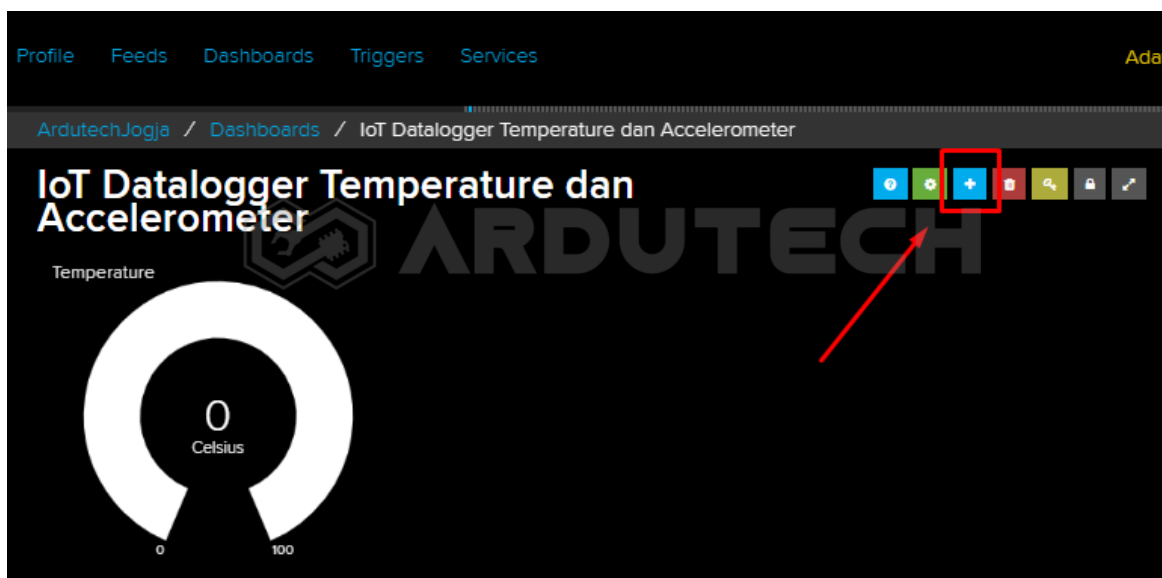
Test Value

45

Isi pada kolom yang tersedia :

- Block Title : Temperature (opsional)
- Gauge Min Value : 0
- Gauge Max Value : 100
- Gauge Width : 50px
- Gauge Label : Celsius

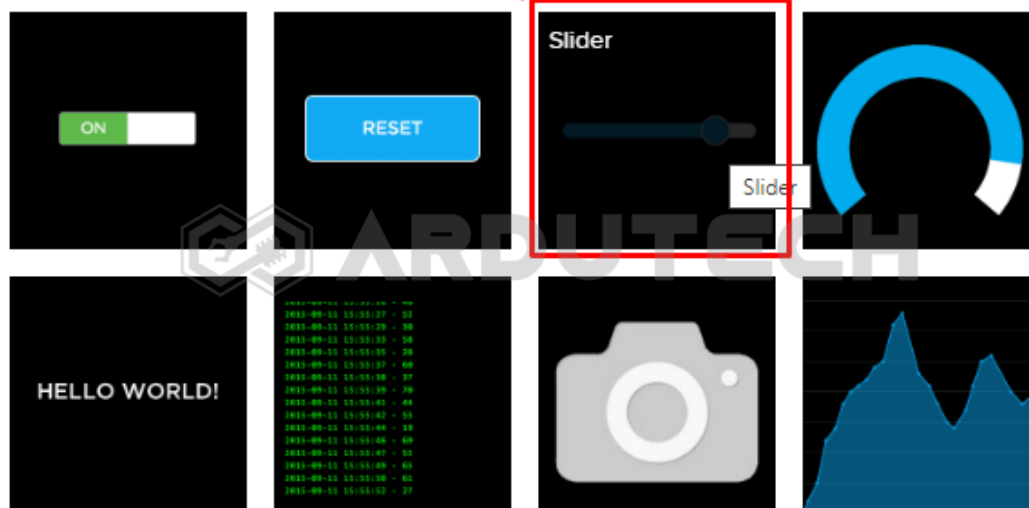
Klik tombol **"Create block"**. Selesai pembuatan antarmuka di Adafruit.IO.



Klik tombol plus "+" lagi untuk menambahkan sebuah **Slider** ke dashboard.

Create a new block

Click on the block you would like to add to your dashboard. You can always come back and switch the block type later if you change your mind.



Selanjutnya muncul tampilan untuk memilih 'Chose feed'. Ini semacam "Topik" dalam komunikasi MQTT. Jadi supaya dikenali oleh 'subscriber' (penerima perintah) maka 'publisher' harus menentukan 'topik' nya. Silakan tulis "**accel_x**" kemudian klik tombol "**Create**".

Choose feed

Slider: The slider works well if you have a range of values you need to send.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Search field:

Group / Feed	Last value	Recorded
☐ Humidity	73.00	about 22 h...
☐ LED	ON3	1 day
☐ temperature	29.60	about 22 h...

Muncul feed accel_x :

Choose feed

Slider: The slider works well if you have a range of values you need to send.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Search field:

Group / Feed	Last value	Recorded
☒ accel_x		less than a ...
☐ Humidity	73.00	about 22 h...
☐ LED	ON3	1 day
☐ temperature	29.60	about 22 h...

Selanjutnya klik “**Next Step** > “ sehingga muncul tampilan **Block settings** :

Block settings

In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

Block Title (optional)

Angle X

Slider Min Value

0

Slider Max Value

360

Slider Step Size

10

Slider Label

(m/s^2)

Decimal Places

4

Number of decimal places to display, defaults to 4.

Block Preview

Angle X

45

Slider

The slider works well if you have a range of values you need to send.

Test Value

45

Published Value

0 bytes

< Previous step

Create block

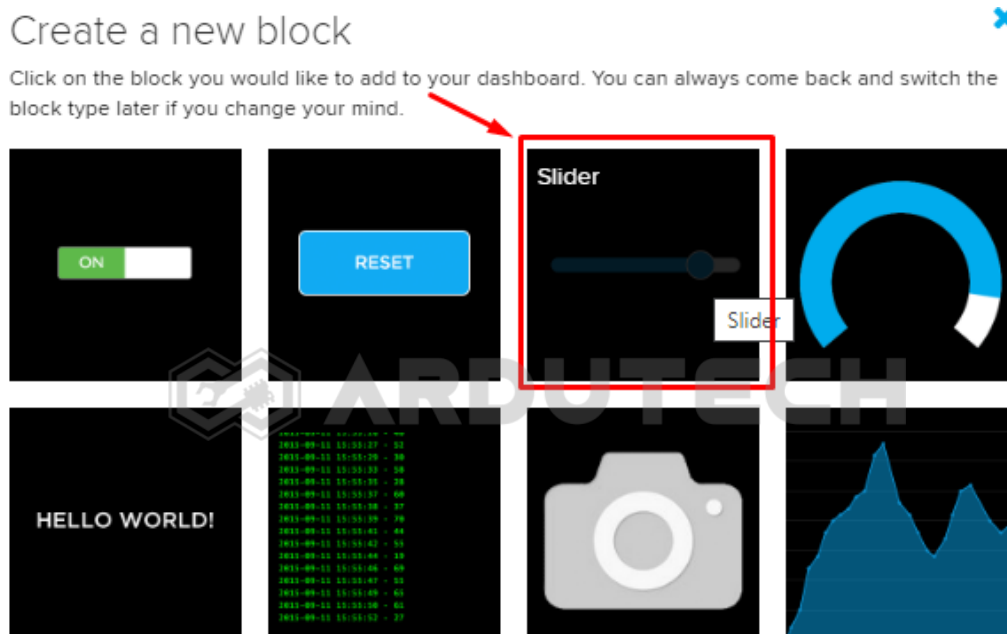
Isi pada kolom yang tersedia :

- Block Title : Angle X (opsional)
- Gauge Min Value : 0
- Gauge Max Value : 360
- Gauge Width : 10
- Gauge Label : (m/s^2)

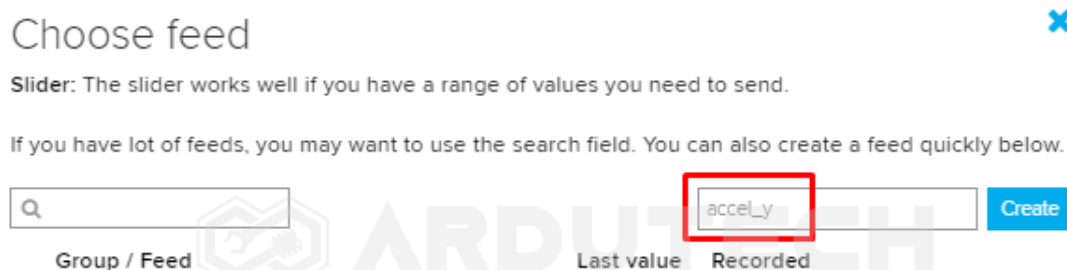
Klik tombol “**Create block**”, sekarang tampilan dashboard menjadi seperti gambar berikut :



Klik tombol plus “+” lagi untuk menambahkan sebuah **Slider** ke dashboard.



Selanjutnya muncul tampilan untuk memilih 'Chose feed'. Ini semacam “Topik” dalam komunikasi MQTT. Jadi supaya dikenali oleh ‘subscriber’ (penerima perintah) maka ‘publisher’ harus menentukan ‘topik’ nya. Silakan tulis “**accel_y**” kemudian klik tombol “**Create**”.



Muncul feed **accel_y** :

Choose feed

Slider: The slider works well if you have a range of values you need to send.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Create

Group / Feed	Last value	Recorded
☐ accel_x	🔒	6 minutes
☒ accel_y	🔒	less than a ...
☐ Humidity	🔒 73.00	about 22 h...
☐ LED	🔒 ON3	1 day
☐ temperature	🔒 29.60	about 22 h...

Previous step

Next step >

Selanjutnya klik “**Next Step >**” sehingga muncul tampilan **Block settings** :

Block settings

In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

Block Title (optional)

Angle Y

Slider Min Value

0

Slider Max Value

360

Slider Step Size

10

Slider Label

(m/s^2)

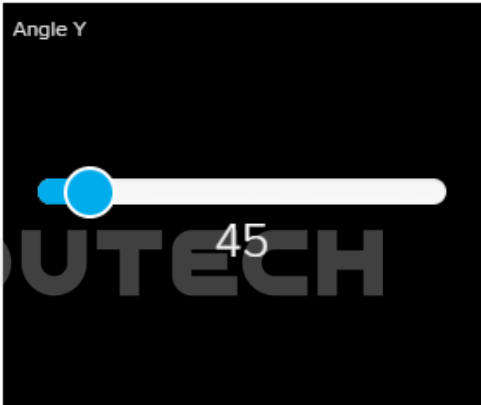
Decimal Places

4

Number of decimal places to display, defaults to 4.

Block Preview

Angle Y



Slider The slider works well if you have a range of values you need to send.

Test Value

45

Published Value

0 bytes

Previous step

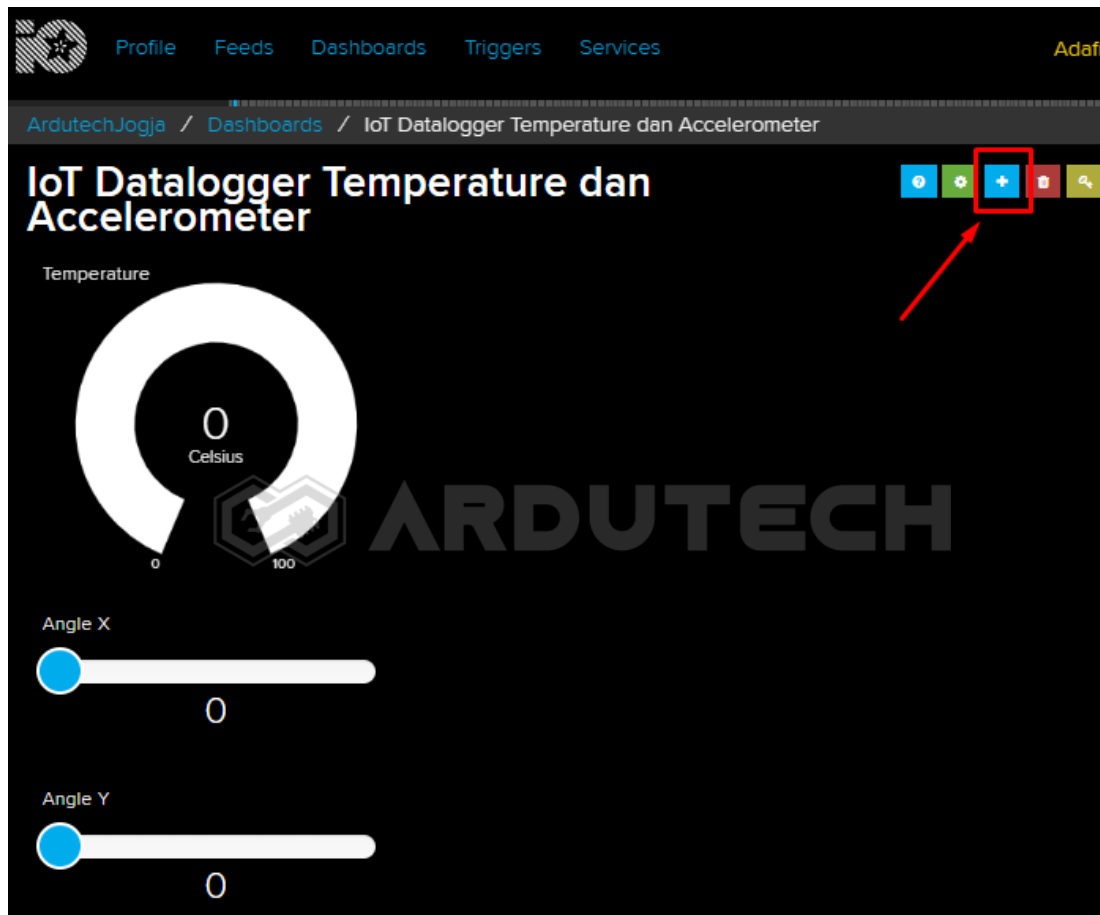
Create block

Isi pada kolom yang tersedia :

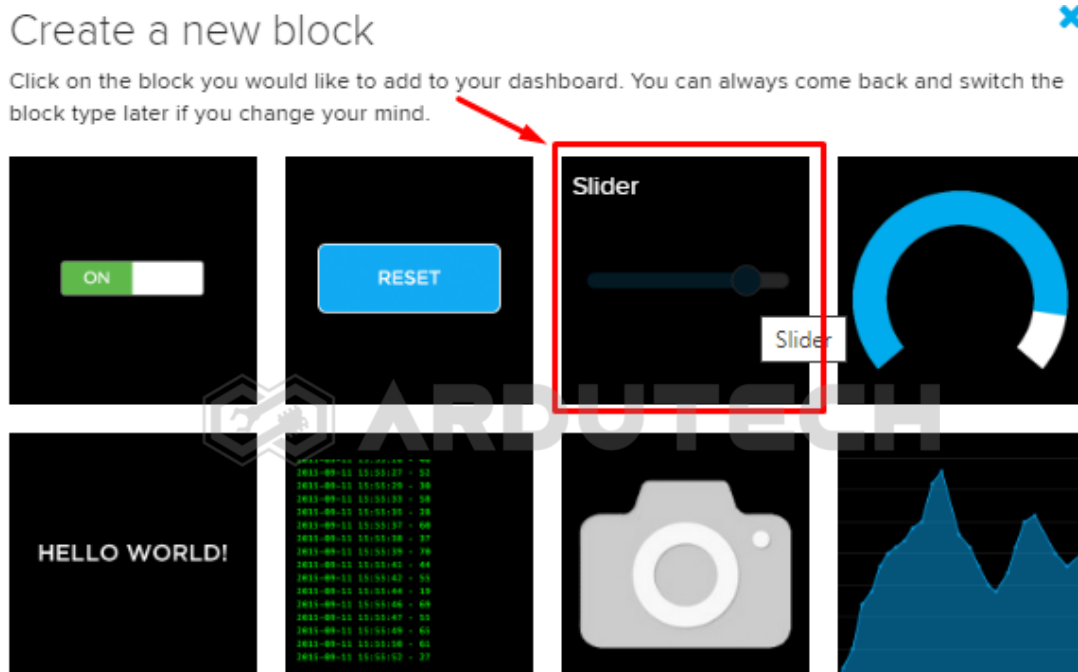
- Block Title : Angle Y (opsional)
- Gauge Min Value : 0

- Gauge Max Value : 360
- Gauge Width : 10
- Gauge Label : (m/s²)

Klik tombol “**Create block**”, sekarang tampilan dashboard menjadi seperti gambar berikut :



Klik tombol plus “+” lagi untuk menambahkan sebuah **Slider** ke dashboard.



Selanjutnya muncul tampilan untuk memilih '**Chose feed**'. Ini semacam "Topik" dalam komunikasi MQTT. Jadi supaya dikenali oleh 'subscriber' (penerima perintah) maka 'publisher' harus menentukan 'topik' nya. Silakan tulis "**accel_z**" kemudian klik tombol "**Create**".

Choose feed

Slider: The slider works well if you have a range of values you need to send.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Search field:

Group / Feed	Last value	Recorded
--------------	------------	----------

Muncul feed **accel_z** :

Choose feed

Slider: The slider works well if you have a range of values you need to send.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Search field:

Group / Feed	Last value	Recorded
☐ accel_x	🔒	10 minutes
☐ accel_y	🔒	4 minutes
☒ accel_z	🔒	less than a ...
☐ Humidity	🔒 73.00	about 22 h...
☐ LED	🔒 ON3	1 day
☐ temperature	🔒 29.60	about 22 h...

Selanjutnya klik "**Next Step >**" sehingga muncul tampilan **Block settings** :

Block settings

In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

Block Title (optional)

Angle Z

Slider Min Value

0

Slider Max Value

360

Slider Step Size

10

Slider Label

(m/s^2)

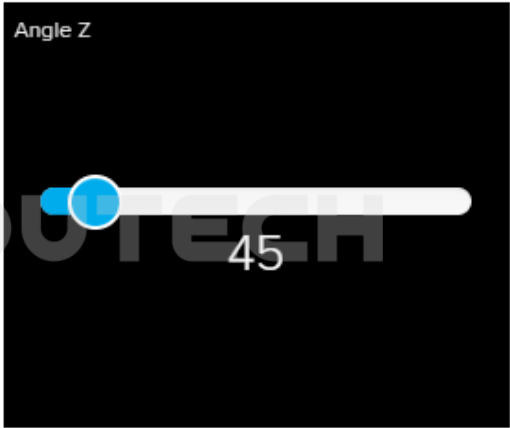
Decimal Places

4

Number of decimal places to display, defaults to 4.

Block Preview

Angle Z



Slider

The slider works well if you have a range of values you need to send.

Test Value

45

Published Value

0 bytes

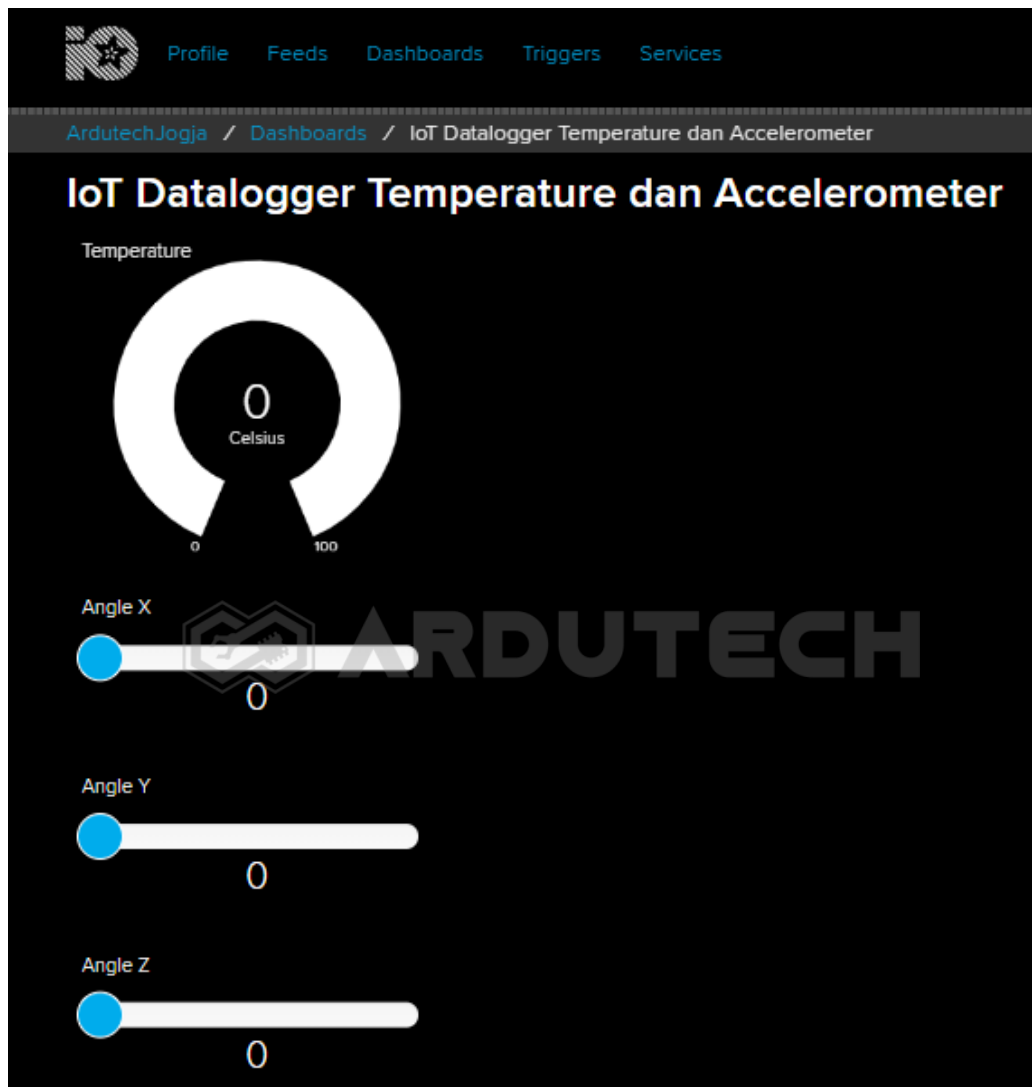
← Previous step

Create block

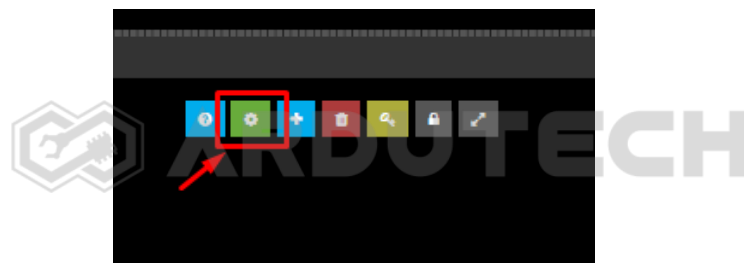
Isi pada kolom yang tersedia :

- Block Title : Angle Z (opsional)
- Gauge Min Value : 0
- Gauge Max Value : 360
- Gauge Width : 10
- Gauge Label : (m/s^2)

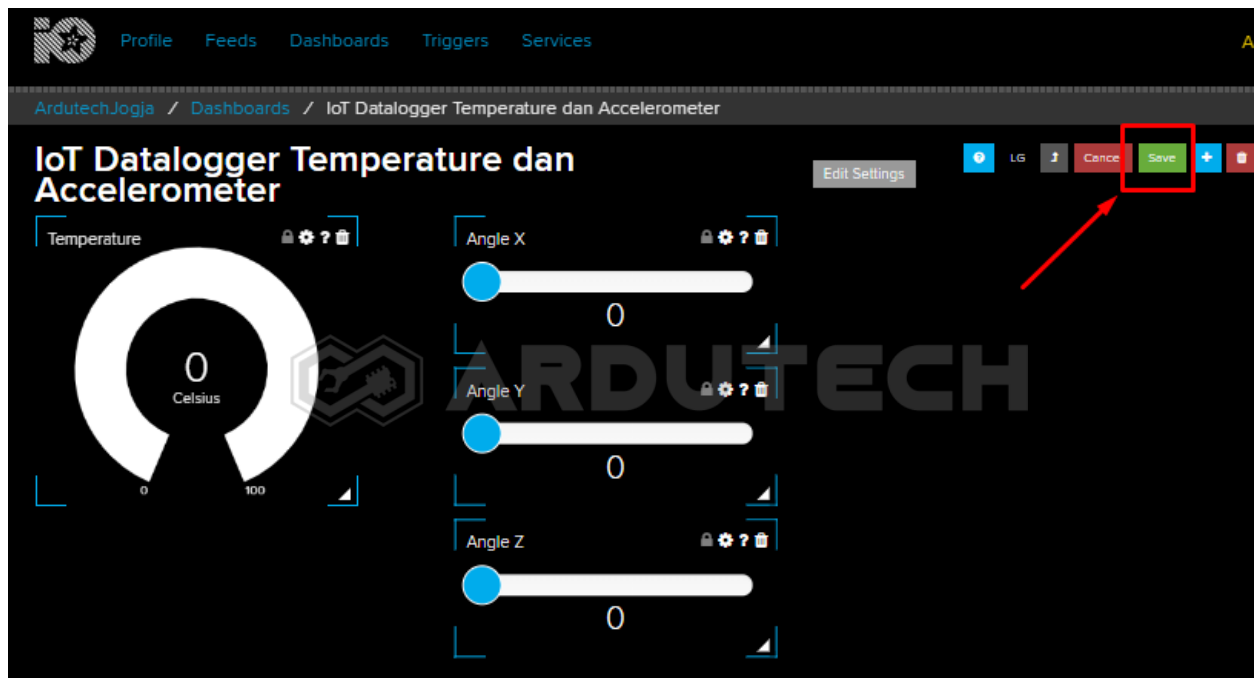
Klik tombol "**Create block**", sekarang tampilan dashboard menjadi seperti gambar berikut :



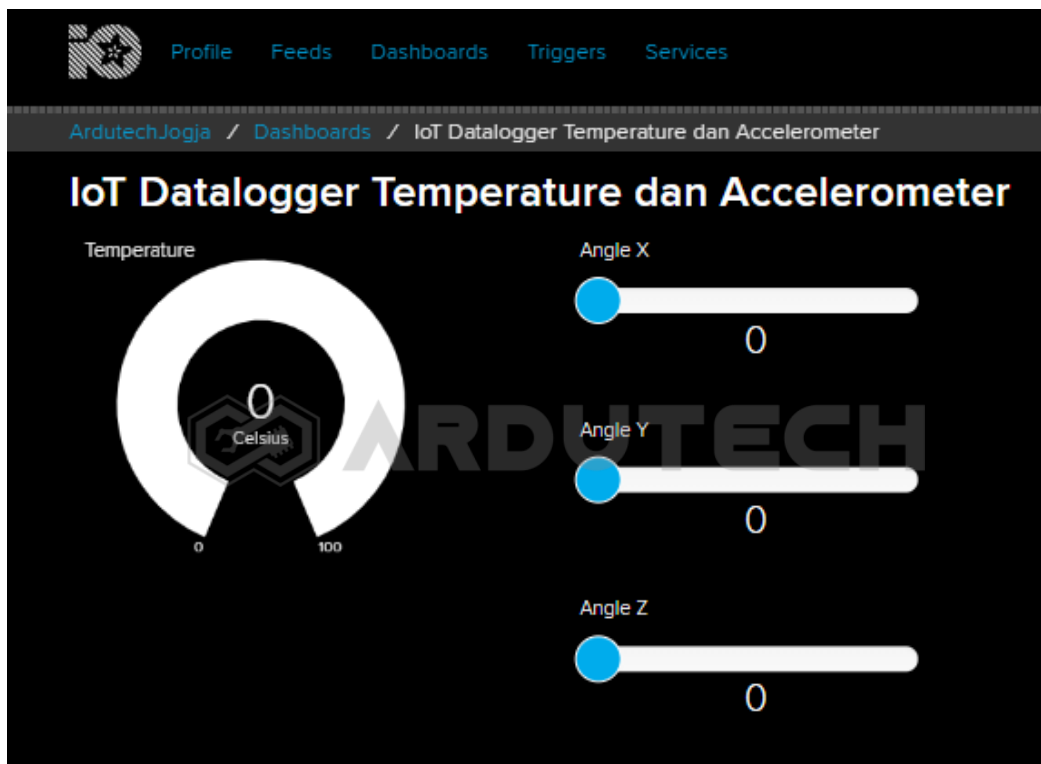
Ok selesai untuk pembuatan dashboardnya, jika ingin mengatur tata letaknya (layout) silakan klik **"Edit dashboard layout "** :



Silakan atur tata letaknya sesuai keinginan anda.



Kalau sudah klik “ Save ”.



Selanjutnya kita perlu kode (token) untuk mengisi variabel username dan active key di program Arduino yang akan dibuat. Klik bagian “**Adafruit IO Key**” di pojok kanan atas.

YOUR ADAFRUIT IO KEY

Your Adafruit IO Key should be kept in a safe place and treated with the same care as your Adafruit username and password. People who have access to your Adafruit IO Key can view all of your data, create new feeds for your account, and manipulate your active feeds.

If you need to regenerate a new Adafruit IO Key, all of your existing programs and scripts will need to be manually changed to the new key.



Username

ArdutechJogja

Active Key

ruu_wUXJ670DJNEIBQWERTY567MNB

REGENERATE KEY

[Hide Code Samples](#)

Arduino

```
#define IO_USERNAME "ArdutechJogja"
#define IO_KEY      "ruu_wUXJ670DJNEIBQWERTY567MNB"
```

Perhatikan untuk **Username** dan **Active Key** ini penting, karena nanti dipakai ketika pemrograman dengan Arduino IDE. Catat/simpan kode tersebut.

Nah untuk pembuatan antarmuka pada Adafruit sudah selesai, sekarang kita siapkan program Arduino IDE-nya.

Program/Source Code

Program pada proyek ini memerlukan library :

- ESP8266WiFi.h
- Adafruit_MQTT.h
- Adafruit_MQTT_Client.h
- Wire.h

Buka/jalankan Arduino IDE kemudian buat lembar kerja baru. Tulis kode program berikut.

```
/******
 * Project Monitoring Accelerometer
 * Board   : NodeMCU ESP8266 V3
 * Input   : Sensor MPU-6050
 * Output  : Blynk
 * 99 Proyek IoT
 * www.ardutech.com
 *****/
#include<Wire.h>
#include <ESP8266WiFi.h>
```

```
#include "Adafruit_MQTT.h"
#include "Adafruit_MQTT_Client.h"

//---GANTI SESUAI DENGAN JARINGAN WIFI
//---HOTSPOT ANDA
#define WLAN_SSID      "ArdutechWiFi"
#define WLAN_PASS      "12345678"
#define AIO_SERVER      "io.adafruit.com"
#define AIO_SERVERPORT  1883           // use 8883 for SSL
//---GANTI SESUAI DENGAN USER NAME ADAFRUIT ANDA
#define AIO_USERNAME    "ArdutechJogja"
//---GANTI SESUAI DENGAN IO KEY ADAFRUIT ANDA
#define AIO_KEY          "aio_mAXJ90DJnEIBVj0IDmG6BIMZqMAW"

const int MPU_addr=0x68;
int16_t AcX,AcY,AcZ;
int16_t temp;
int minVal=265;
int maxVal=402;

double x;
double y;
double z;

WiFiClient client;

Adafruit_MQTT_Client mqtt(&client, AIO_SERVER, AIO_SERVERPORT,
AIO_USERNAME, AIO_KEY);

Adafruit_MQTT_Publish temperature = Adafruit_MQTT_Publish(&mqtt,
AIO_USERNAME "/feeds/temperature");

Adafruit_MQTT_Publish angle_x = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME
"/feeds/accel_x");
```

```
Adafruit_MQTT_Publish angle_y = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME
"/feeds/accel_y");

Adafruit_MQTT_Publish angle_z = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME
"/feeds/accel_z");

void MQTT_connect();

//=====

void setup(){
  Wire.begin();
  Wire.beginTransmission(MPU_addr);
  Wire.write(0x6B);
  Wire.write(0);
  Wire.endTransmission(true);
  Serial.begin(9600);
  Serial.println(); Serial.println();
  Serial.print("Connecting to ");
  Serial.println(WLAN_SSID);

  WiFi.begin(WLAN_SSID, WLAN_PASS);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println();
  Serial.println("WiFi connected");
  delay(5000);
}

//=====

void loop(){
  MQTT_connect();
  Wire.beginTransmission(MPU_addr);
  Wire.write(0x3B);
```



```
Wire.endTransmission(false);

Wire.requestFrom(MPU_addr,14,true);

AcX=Wire.read()<<8|Wire.read();
AcY=Wire.read()<<8|Wire.read();
AcZ=Wire.read()<<8|Wire.read();

temp = Wire.read()<<8 | Wire.read(); //

int xAng = map(AcX,minVal,maxVal,-90,90);
int yAng = map(AcY,minVal,maxVal,-90,90);
int zAng = map(AcZ,minVal,maxVal,-90,90);

x= RAD_TO_DEG * (atan2(-yAng, -zAng)+PI);
y= RAD_TO_DEG * (atan2(-xAng, -zAng)+PI);
z= RAD_TO_DEG * (atan2(-yAng, -xAng)+PI);
temp=temp/340.00+36.53;

if (! temperature.publish(temp)) {
    Serial.println(F("Publish Temperature Failed"));
} else {
    Serial.println(F("Publish Temperature OK!"));
}

if (! angle_x.publish(x)) {
    Serial.println(F("Publish Angle X Failed"));
} else {
    Serial.println(F("Publish Angle X OK!"));
}

if (! angle_y.publish(y)) {
    Serial.println(F("Publish Angle Y Failed"));
} else {
    Serial.println(F("Publish Angle Y OK!"));
}
```

```

if (! angle_z.publish(z)) {
    Serial.println(F("Publish Angle Z Failed"));
} else {
    Serial.println(F("Publish Angle Z OK!"));
}

Serial.print("Temperature= ");
Serial.println(temp);

Serial.print("AngleX= ");
Serial.println(x);

Serial.print("AngleY= ");
Serial.println(y);

Serial.print("AngleZ= ");
Serial.println(z);
Serial.println("-----");
delay(5000);
}

//=====
void MQTT_connect() {
    int8_t ret;

    // Stop if already connected.
    if (mqtt.connected()) {
        return;
    }

    Serial.print("Connecting to MQTT... ");

```

```
uint8_t retries = 3;

while ((ret = mqtt.connect()) != 0) { // connect will return 0 for
connected

    Serial.println(mqtt.connectErrorString(ret));
    Serial.println("Retrying MQTT connection in 5 seconds...");
    mqtt.disconnect();
    delay(5000); // wait 5 seconds
    retries--;
    if (retries == 0) {
        while (1);
    }
}

Serial.println("MQTT Connected!");
}
```

PERHATIKAN !

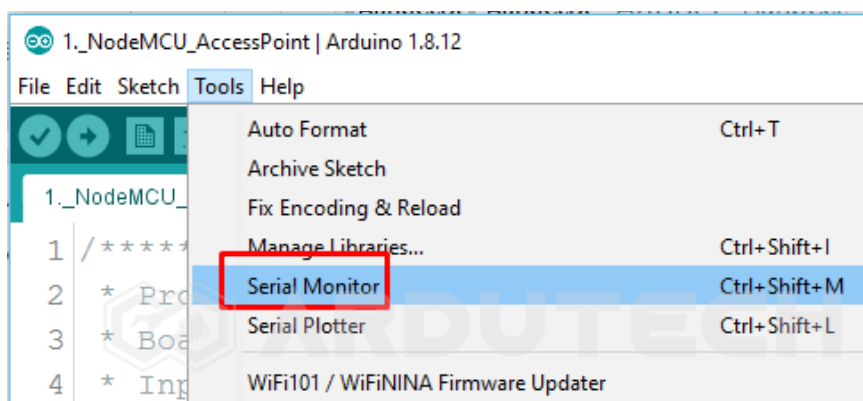
Ganti/sesuaikan variabel berikut :

- Nama jaringan WiFi/hotspot : **WLAN_SSID**
- Password jaringan WiFi/hotspot : **WLAN_PASS**
- Username Adafruit IO : **AIO_USERNAME**
- Active Key Adafruit IO : **AIO_KEY**

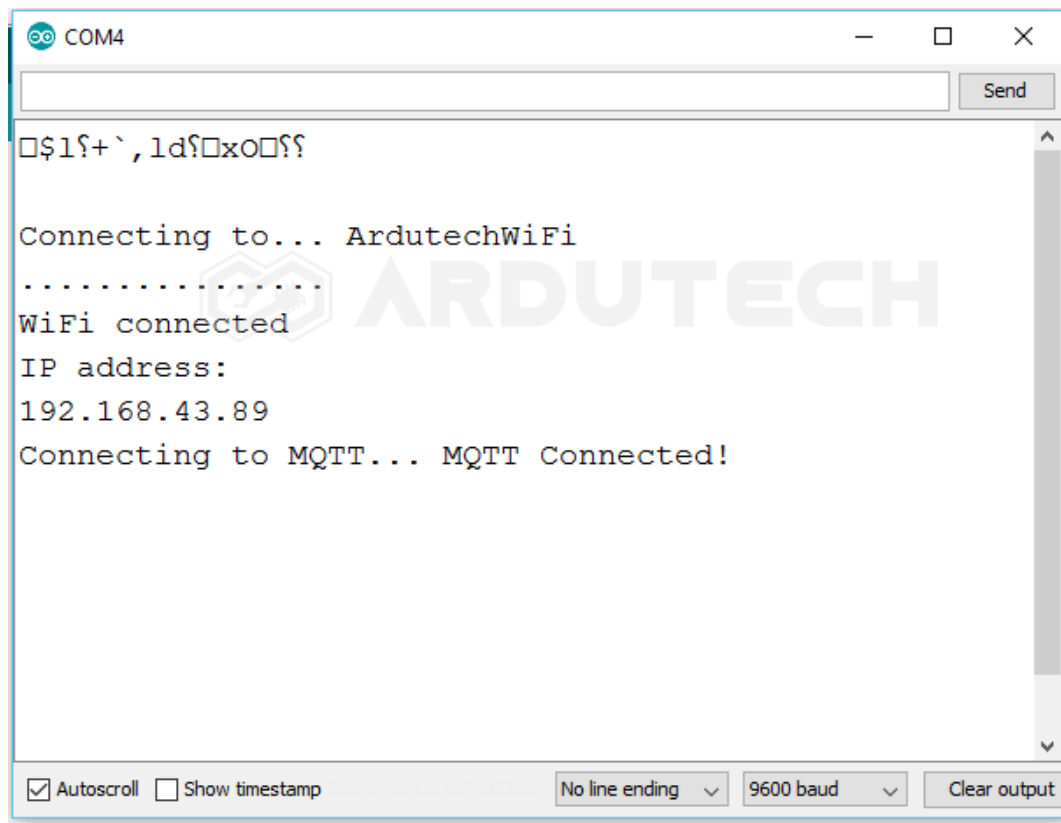
Simpan (**Save**) kemudian **Upload**. Pastikan tidak ada error, jika masih ada silakan cek penulisan dll kemudian perbaiki. (Program ini sudah diuji langsung dan sudah berjalan tanpa ada error)

Jalannya Alat

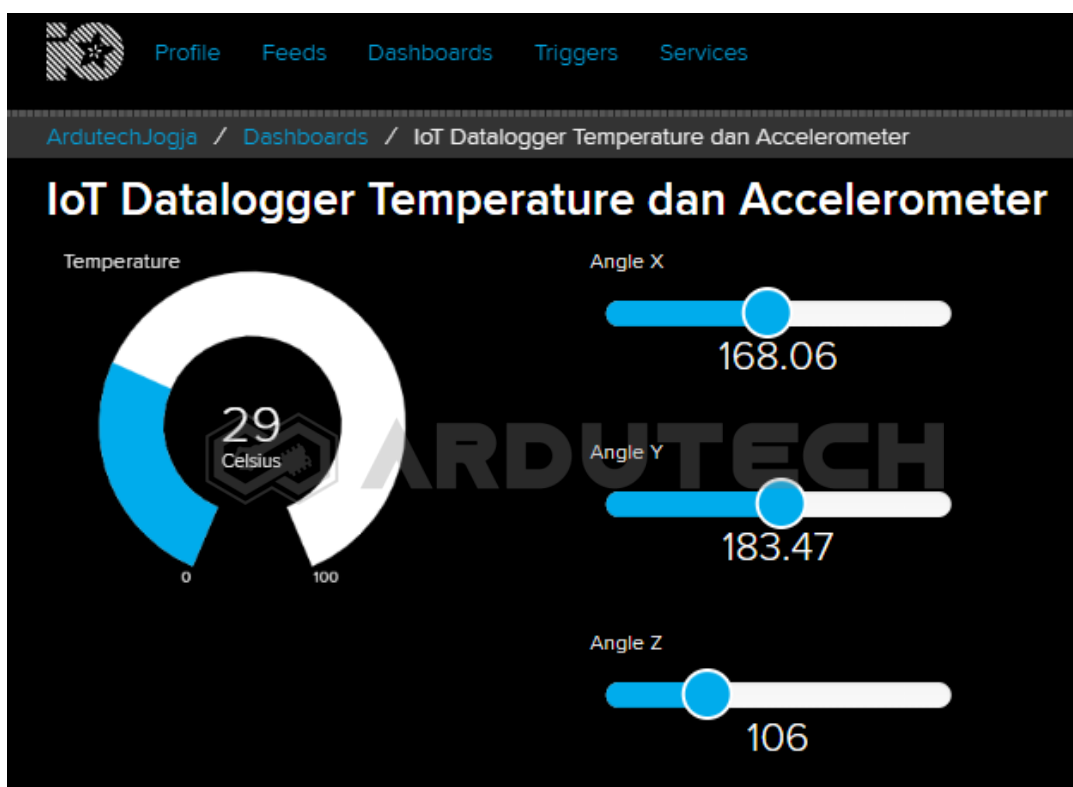
Siapkan jaringan WiFi/hotspot atau tethering dari HP dan aktifkan. Buka Serial Monitor, dari menu **Tools → Serial Monitor** kemudian seting baudrate pada nilai 9600.



Jika belum muncul tekan tombol reset (RST) pada board NodeMCU sehingga muncul tampilan sebagai berikut :



Setelah berhasil terhubung dengan jaringan WiFi sekarang kita kembali ke tampilan di Adafruit. Perhatikan nilai pada Gauge menunjukkan hasil pembacaan temperatur /suhu sensor MPU-6050 dan slider menunjukkan hasil pembacaan acceleromernya.





Beri perubahan suhu pada sensor MPU-6050 kemudian perhatikan perubahan pada nilai yang tampil di dashboard Adafruit .

Selamat berkarya , semoga lancar dan bermanfaat.

Ardutech – *“Sahabat Inovasi Anda”*